Cyclic Incremental Latin $n \times n$ Squares encoding Generalized (n, c)-Coins Squares for $0 \leq c \leq n$

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Abstract

For every natural number $n \notin \{0, 2, 3, 4, 6\}$, there exists a permutation $q : \operatorname{Fin} n \to \operatorname{Fin} n$ such that $L_{i,j} = (i + q(j)) \bmod n$ is a *Cyclic Incremental Latin Square*, encoding (n, c)-Coins Squares for $0 \leq c \leq n$.

An (n, c)-Coins Square for $c = 1$ corresponds to a solution of the n-Queens Problem.

We have an existence proof in `Lean` and construction code in `Haskell`, `Julia` and `Lean`.

See e.g. MO#510135, PSE#138939, MO#516225 and GitHub.

The proof uses the *Injective-Majorization Principle* that, in combination with the *Pigeonhole Principle*, replaces intensive checking inequalities on each diagonal with the existence of a single function $f(i, j) \leq L_{i,j}$ that is injective on every diagonal.

The proof uses four families covering almost all natural number modulo 6 classes:

- for $n \equiv 1, 5 \pmod 6$ (with $n \geq 1$)
- for $n \equiv 2, 4 \pmod 6$ (with $n \geq 10$)
- for $n \equiv 0 \pmod 6$ (with $n \geq 12$)
- for $n \equiv 3 \pmod 6$ (with $n \geq 27$)

For all four families a Cyclic Incremental Latin Square can be constructed. Constructing a Cyclic Incremental Latin Square for the four uncovered natural numbers, $n \in \{8, 9, 15, 21\}$, is handled separately. Likewise, constructing a (non-Cyclic) Incremental Latin Square for the two remaining natural numbers, $n \in \{4, 6\}$, is also handled separately.

1 Cyclic Incremental Latin Squares

1.1 An Example: The 8×8 Incremental Square and its $(8, c)$-Coins Squares

As noted in the abstract, $n = 8$ is one of the four individual cases not covered by the four modulo-6 families. Anyway, using the permutation $q = (0, 4, 1, 3, 6, 2, 7, 5)$, $L_{i,j} = (i + q(j)) \bmod 8$ produces the following 8×8 Cyclic Incremental Latin Square:

$$L = \begin{pmatrix} 0 & 4 & 1 & 3 & 6 & 2 & 7 & 5 \\ 1 & 5 & 2 & 4 & 7 & 3 & 0 & 6 \\ 2 & 6 & 3 & 5 & 0 & 4 & 1 & 7 \\ 3 & 7 & 4 & 6 & 1 & 5 & 2 & 0 \\ 4 & 0 & 5 & 7 & 2 & 6 & 3 & 1 \\ 5 & 1 & 6 & 0 & 3 & 7 & 4 & 2 \\ 6 & 2 & 7 & 1 & 4 & 0 & 5 & 3 \\ 7 & 3 & 0 & 2 & 5 & 1 & 6 & 4 \end{pmatrix}.$$

For every threshold $c \in \{1, \ldots, 8\}$, placing a coin (denoted by $\bullet$) in cell (i, j) if and only if $L_{i,j} < c$ yields the $(8, c)$-Coins Square S_c. Each square contains exactly $8c$ coins and has at most c coins along every row, column, and diagonal. Figure 1 displays all eight nested solutions $S_1 \subset S_2 \subset \cdots \subset S_8$, where $(8, 1)$ is a classic 8-Queens solution.

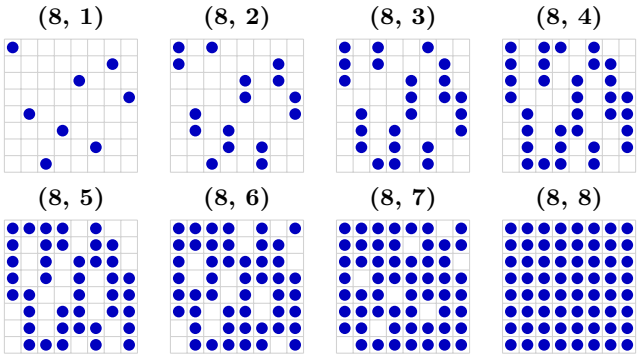


Figure 1: The eight nested $(8, c)$-Coins Squares generated by $L_{i,j} < c$ for $c = 1, \ldots, 8$. Coins ($\bullet$) are added incrementally without moving previously placed coins.

1.2 From (n, c)-Coins Squares to Cyclic Incremental Latin Squares

(n, c)-Coins Squares are subsets $S_c \subseteq \{0, \ldots, n-1\}^2$ of nc cells of an $n \times n$ square such that every row, column, sum-diagonal $(i + j = s)$, and difference-diagonal $(i - j = d)$ contains at most c cells of S_c.

Clearly, for $c = 1$, they correspond to solutions of the n-Queens Problem.

The construction of Le, Li and Wang [1] constructs S_c's separately for each c. In contrast, an *Incremental Latin Square* encodes the entire nested sequence of solutions $S_1 \subset S_2 \subset \cdots \subset S_n$ simultaneously into a single board $L_{i,j}$ by setting $S_c = \{(i, j) \mid L_{i,j} \leq c\}$.

1.3 The Conjecture

Every permutation $q : \mathrm{Fin}\, n \to \mathrm{Fin}\, n$ defines a *Cyclic Latin Square*:

$$L_{i,j} = (i + q(j)) \bmod n \qquad (0 \leq i, j < n).$$

The conjecture is that for all $n > 0$ except $n \in \{2, 3, 4, 6\}$, there exists a column permutation q such that $L_{i,j} = (i + q(j)) \bmod n$ is a Cyclic Incremental Latin Square. For $n \in \{2, 3\}$, the n-queens problem has no solution, and, for $n \in \{4, 6\}$, exhaustive finite search confirms that no Cyclic Incremental Latin Square satisfies the diagonal bounds.

Theorem 1.1 (Cyclic Incremental Latin Square Conjecture)**.** *For every integer $n > 0$ and $n \notin \{2, 3, 4, 6\}$, there exists a bijection $q : \mathrm{Fin}\, n \to \mathrm{Fin}\, n$ such that the Cyclic Incremental Latin Square*

$$L_{i,j} = (i + q(j)) \bmod n$$

satisfies:

(i) For every $i \in \mathrm{Fin}\, n$, the row map $j \mapsto L_{i,j}$ is bijective.

(ii) For every $j \in \mathrm{Fin}\, n$, the column map $i \mapsto L_{i,j}$ is bijective.

(iii) For every $0 \leq k < n$ and every $s \in \mathbb{N}$,

$$|\{(i, j) \in \mathrm{Fin}\, n \times \mathrm{Fin}\, n \mid i + j = s \ \wedge \ L_{i,j} \leq k\}| \leq k + 1.$$

(iv) For every $0 \leq k < n$ and every $d \in \mathbb{Z}$,

$$|\{(i, j) \in \mathrm{Fin}\, n \times \mathrm{Fin}\, n \mid i - j = d \ \wedge \ L_{i,j} \leq k\}| \leq k + 1.$$

2 Injective Majorization

Instead of direct cardinality counting of

$$|\{(i,j) \in \operatorname{Fin} n \times \operatorname{Fin} n \mid i+j = s \ \wedge \ L_{i,j} \leq k\}| \leq k+1$$

and

$$|\{(i,j) \in \operatorname{Fin} n \times \operatorname{Fin} n \mid i-j = d \ \wedge \ L_{i,j} \leq k\}| \leq k+1,$$

which would computationally explode for large n, we use a combinatorial principle that reduces the problem for *all* thresholds $0 \leq k < n$ simultaneously to constructing a single dominated injection.

2.1 The Injective-Majorization Principle

Let $D \subseteq \operatorname{Fin} n \times \operatorname{Fin} n$ be any diagonal (either a sum-diagonal $i+j=s$ or a difference-diagonal $i-j=d$), and let $v(i,j) = (i+q(j)) \bmod n \in \{0,\ldots,n-1\}$ be the board value at cell (i,j).

Lemma 2.1 (Injective-Majorization Principle)**.** *If there exists a function $f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ such that:*
(1) Majorization: $f(i,j) \leq v(i,j)$ for all $0 \leq i,j < n$,
(2) Diagonal Injectivity: f is injective along D.
Then for every threshold $k \in \mathbb{N}$, the set $D_{\leq k} = \{(i,j) \in D \mid v(i,j) \leq k\}$ has cardinality at most $k+1$.

Proof. For every cell $(i,j) \in D_{\leq k}$, condition (1) gives

$$0 \leq f(i,j) \leq v(i,j) \leq k,$$

so $f(i,j) \in \{0,\ldots,k\}$. By condition (2), the restriction of f to $D_{\leq k}$ is an injective map into $\{0,\ldots,k\}$. By the Pigeonhole Principle,

$$|D_{\leq k}| \leq |\{0,\ldots,k\}| = k+1. \qquad \square$$

2.2 Why Vacant Lower Slots Always Exist

Along a diagonal D, the board values $v(i,j)$ are *almost* distinct, except for a small number of collision pairs $(i_1,j_1),(i_2,j_2) \in D$ where $v(i_1,j_1) = v(i_2,j_2) = v$. To satisfy Lemma 2.1, one of the two cells keeps its value $f(i_1,j_1) = v$, while the duplicate cell is reassigned to a strictly smaller value $f(i_2,j_2) < v$ that does not appear anywhere else along D. The reassigned cell *steals* a slot in $\{0,\ldots,v-1\}$ that was left vacant on D.

A natural and fundamental question arises:

> *Why precisely is there always guaranteed to be an unoccupied value strictly less than v available to be stolen?*

The existence of such vacant lower slots is not accidental; it is guaranteed by the arithmetic and geometric structure of the column permutation $q(j)$:

(a) *Elevated Baselines and Block Jumps (The Vacant Lower Shadow):*
Stepping along a diagonal changes j by $+1$, and changes i by -1 (along $i+j=s$) or $+1$ (along $i-j=d$). Within a linear block $q(j) = a \cdot j + b$, the unreduced value expression advances with constant slope:

$$\Delta_{\text{sum}} = a-1, \qquad \Delta_{\text{diff}} = a+1.$$

In Family 2 ($n \geq 10$ even, $3 \nmid n$), each block has slope $a=2$, so $\Delta_{\text{sum}} = +1$. Within each half, the sequence is strictly increasing without any internal collisions. The first block ($j_1 < n/2$) covers cells with high row indices $i_1 = s - j_1$, and starts at an elevated baseline $q(j_1) \geq n/2+1$. Its values along the diagonal start high above 0. The second block ($j_2 \geq n/2$) has a parity jump of $+3$ and lower row indices i_2, causing it to wrap around n and hit the same value v as cell (i_1,j_1). Crucially, because the diagonal entered the duplicate value v while skipping the lower numbers $\{0,\ldots,\text{corner size}-1\}$, *none of the cells on the diagonal ever visited those lowest values.* They form a vacant "shadow" $\{0,\ldots,v-1\}$. Reassigning the duplicate cell (i_2,j_2) to $j_2-(n-3) \in \{0,1,2\}$ steals one of these untouched slots, with $f(i_2,j_2) < v$.

(b) *Modular Wrap-Around and Empty Modulo-3 Residue Classes:*
In Family 3 ($n \geq 12$, $6 \mid n$), the column permutation has slope $a = 2$. Along a difference-diagonal $i - j = d$, the progression has step $\Delta_{\text{diff}} = 2 + 1 = 3$. Because $3 \mid n$, stepping by $+3$ confines all cells within a linear block to a *single residue class modulo* 3 ($d + 2 \pmod 3$ for Block 1, and $d \pmod 3$ for Block 2). Consequently, *the third residue class* $d + 1 \pmod 3$ *is left completely empty on the entire diagonal* (one-third of all values $\{0, \ldots, n-1\}$ are unvisited). When j advances by $k = n/3$, the linear term advances by $3(n/3) = n$. Modulo n, this wraps around, causing a pure *modular wrap-around at* n that repeats the exact same value v from k steps earlier. Because the parallel residue class $d+1 \pmod 3$ is entirely vacant, shifting the wrapped cell downward by -1 or -2 lands it directly into an empty slot in $\{0, \ldots, v-1\}$.

(c) *Pigeonhole Conservation:*
On any diagonal of length $|D|$, if a value v appears twice, at most $|D| - 1$ distinct values appear among the $|D|$ cells. Whenever the sequence steps into $\{0, \ldots, v\}$ with stride $\Delta \geq 1$ from a non-zero offset, the number of cells on D taking values $\leq v$ is strictly less than $v + 1$. By the Pigeonhole Principle, the set $\{0, \ldots, v-1\}$ cannot be full, guaranteeing the existence of vacant slots below v.

2.3 Visualization of Four Concrete Examples of Diagonal Slot Stealing

We present four concrete examples from the Lean formalization: two for sum-diagonals and two for difference-diagonals, illustrating piecewise jumps and modular wrap-around. In each grid, off-diagonal values and the grid lines are displayed in gray, diagonal cells are highlighted in blue, the cell that keeps its original value is highlighted in orange ($\mathbf{v}$), and the duplicate cell that steals a lower vacant slot is highlighted in red ($\mathbf{v} \downarrow \mathbf{f}$).

Example 2.2 (Sum-Diagonal Piecewise Jump: $n = 10$, $s = 7$). For $n = 10$, $q = (6, 8, 0, 2, 4 \mid 9, 1, 3, 5, 7)$ has two linear halves ($j < 5$ and $j \geq 5$). Sum-diagonal $i + j = 7$ has 8 cells. Cells $(7, 0)$ in Block 1 and $(0, 7)$ in Block 2 both evaluate to $v = 3$:

$$v(7,0) = (7 + q(0)) \bmod 10 = (7+6) \bmod 10 = 3, \qquad v(0,7) = (0 + q(7)) \bmod 10 = (0+3) \bmod 10 = 3.$$

The collision is caused by the jump across the two halves ($j_2 - j_1 = 7 - 0 = n - 3$). The values present on the diagonal are $\{1, 2, \mathbf{3}, 4, 5, 6, 7\}$. The value $\mathbf{0}$ is *vacant* on the entire diagonal, and $0 < 3$. The majorant $f_{2,\text{sum}}$ keeps $f(7, 0) = 3$ and reassigns the top-right corner cell:

$$f(0,7) = j - (n-3) = 7 - 7 = 0 < 3.$$

The resulting values of f along the diagonal are $\{0, 1, 2, 3, 4, 5, 6, 7\}$, which are all distinct.

Example 2.3 (Sum-Diagonal Multiple Collisions: $n = 10$, $s = 10$). On diagonal $i + j = 10$ ($n = 10$), two simultaneous collisions occur:

- $(9, 1)$ and $(2, 8)$ both have $v = 7$ (with $j_2 - j_1 = 8 - 1 = n - 3$).
- $(8, 2)$ and $(1, 9)$ both have $v = 8$ (with $j_2 - j_1 = 9 - 2 = n - 3$).

The values present on the diagonal are $\{0, 4, 5, 6, \mathbf{7}, \mathbf{8}, 9\}$. The entire range $\{\mathbf{1}, \mathbf{2}, 3\}$ is *vacant*! Cells $(9, 1)$ and $(8, 2)$ keep their values 7 and 8, while the corner cells steal:

$$f(2,8) = 8 - 7 = 1 < 7, \qquad f(1,9) = 9 - 7 = 2 < 8.$$

All nine values of f are now distinct: $\{0, 1, 2, 4, 5, 6, 7, 8, 9\}$.

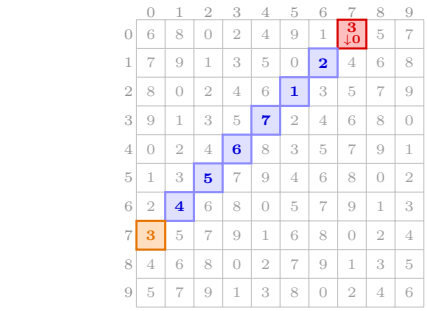
Example 2.4 (Difference-Diagonal Parity Jump: $n = 10$, $d = 0$). Along the main difference-diagonal $i - j = 0$ ($i = j$) for $n = 10$:

$$v(0,0) = (0 + q(0)) \bmod 10 = 6, \qquad v(9,9) = (9 + q(9)) \bmod 10 = (9+7) \bmod 10 = 6.$$

This single collision across the entire board occurs between the first cell of Block 1 ($j = 0$) and the last cell of Block 2 ($j = n - 1 = 9$). The values along $d = 0$ are $\{0, 2, 3, 4, 5, \mathbf{6}, 7, 8, 9\}$. The value $\mathbf{1}$ is *vacant* on the main diagonal, and $1 < 6$. The majorant $f_{2,\text{diff}}$ leaves $(0, 0)$ at 6 and shifts the bottom-right corner cell:

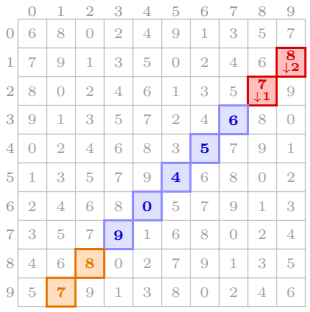
$$f(9,9) = 1 < 6.$$

The resulting values of f form the complete permutation $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.



Cell (i,j)	$v(i,j)$	$f(i,j)$	Status
$(7,0)$	3	3	Keeps value
$(6,1)$	4	4	Distinct
$(5,2)$	5	5	Distinct
$(4,3)$	6	6	Distinct
$(3,4)$	7	7	Distinct
$(2,5)$	1	1	Distinct
$(1,6)$	2	2	Distinct
$(0,7)$	3	**0**	**Steals vacant 0 (<3)**

(a) Sum-diagonal $s = 7$ on $n = 10$.



Cell (i,j)	$v(i,j)$	$f(i,j)$	Status
$(9,1)$	7	7	Keeps value
$(8,2)$	8	8	Keeps value
$(7,3)$	9	9	Distinct
$(6,4)$	0	0	Distinct
$(5,5)$	4	4	Distinct
$(4,6)$	5	5	Distinct
$(3,7)$	6	6	Distinct
$(2,8)$	7	**1**	**Steals vacant 1 (<7)**
$(1,9)$	8	**2**	**Steals vacant 2 (<8)**

(b) Sum-diagonal $s = 10$ on $n = 10$.

Figure 2: Sum-diagonal collisions caused by piecewise block jumps in $q(j)$ for $n = 10$. Off-diagonal values and grid lines are shown in gray. The red cells steal unoccupied slots $0, 1, 2$ in the top-right corner.

Example 2.5 (Difference-Diagonal Pure Modular Wrap-Around: $n = 12$, $d = -4$). For $n = 12$ ($m = 6, k = n/3 = 4$), the second main block of q_3 has $q(j) = 2(j-5)+1$ for $j \in \{5, \ldots, 10\}$. Along the difference-diagonal $i - j = -4$ ($i = j - 4$), stepping $j \mapsto j + 1$ increases $i + q(j)$ by $1 + 2 = +3$. As j advances from 5 to 8, the values advance by $+3$:

$$v(1,5) = 2, \quad v(2,6) = 5, \quad v(3,7) = 8, \quad v(4,8) = 11.$$

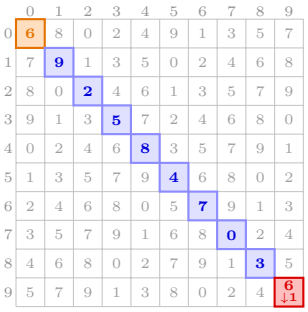
At $j = 9 = 5 + k$, advancing by $k = n/3$ adds $+3 \times 4 = +12 = n$. Modulo 12, the value wraps around:

$$v(5,9) = (5 + q(9)) \bmod 12 = (5 + 9) \bmod 12 = 14 \bmod 12 = 2 = v(1,5)!$$

Similarly, at $j = 10$, $v(6,10) = (6 + 11) \bmod 12 = 17 \bmod 12 = 5 = v(2,6)!$ Both collisions are pure *modular wrap-around at n* within the same linear progression. Because the progression has step $+3$, all cells in this block belong to 2 (mod 3). The boundary cells $(0,4)$ and $(7,11)$ have values 0 and 9 (0 (mod 3)). Thus, *the entire residue class* 1 (mod 3) *($\{1,4,7,10\}$) is completely vacant on the diagonal!* The majorant $f_{3,\text{diff}}$ shifts the wrapped cells into this empty residue class:

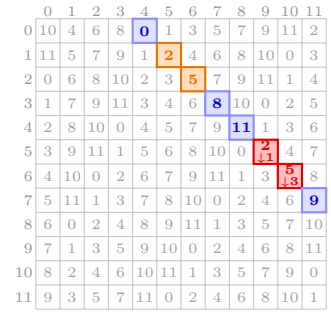
$$f(5,9) = v(5,9) - 1 = 2 - 1 = 1 < 2, \qquad f(6,10) = v(6,10) - 2 = 5 - 2 = 3 < 5.$$

The resulting values of f are $\{0, 1, 2, 3, 5, 8, 9, 11\}$, which are strictly distinct and satisfy $f \le v$.



Cell (i,j)	$v(i,j)$	$f(i,j)$	Status
$(0,0)$	6	6	Keeps value
$(1,1)$	9	9	Distinct
$(2,2)$	2	2	Distinct
$(3,3)$	5	5	Distinct
$(4,4)$	8	8	Distinct
$(5,5)$	4	4	Distinct
$(6,6)$	7	7	Distinct
$(7,7)$	0	0	Distinct
$(8,8)$	3	3	Distinct
$(9,9)$	6	**1**	**Steals vacant 1** (< 6)

(a) Difference-diagonal $d = 0$ on $n = 10$.



Cell (i,j)	$v(i,j)$	$f(i,j)$	Status
$(0,4)$	0	0	Boundary pin
$(1,5)$	2	2	Keeps value
$(2,6)$	5	5	Keeps value
$(3,7)$	8	8	Distinct
$(4,8)$	11	11	Distinct
$(5,9)$	2	**1**	**Steals vacant 1** (< 2)
$(6,10)$	5	**3**	**Steals vacant 3** (< 5)
$(7,11)$	9	9	Boundary pin

(b) Difference-diagonal $d = -4$ on $n = 12$.

Figure 3: Difference-diagonal collisions and reassignment. Off-diagonal values and grid lines are shown in gray. (a) Parity jump between $(0,0)$ and $(9,9)$ for $n = 10$. (b) Pure modular wrap-around at $n = 12$ inside a linear block with step $+3$, where wrapped cells land in the vacant residue class 1 (mod 3).

References

[1] Mao-Hua Le, Wen Li, and Edward T. Wang, *A generalization of the n-queen problem*, Journal of Systems Science and Mathematical Sciences **3** (1990), no. 2, 183–192, https://sysmath.cjoe.ac.cn/jssc/CN/Y1990/V3/I2/183.